

Fused MoE Inference on Blackwell: A Pure Triton Approach to DeepSeek-V3 Expert Dispatch

Jiayao Zhang, Yujiao Liang
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Abstract

We present a pure Triton implementation of the DeepSeek-V3 MoE kernel targeting the NVIDIA B200 (Blackwell, sm_100a) GPU. Our 6-stage pipeline exploits FP8 native tensor core dot products for $2\times$ throughput in GEMM1, a non-atomic two-pass GEMM2-then-reduce architecture, FP16 intermediate buffers with scale-and-cast compensation, and multi-level bucket specialization across four BLOCK_M tiers. On 19 real-trace workloads ($T=1$ to $T=14,107$), the system achieves 19/19 correctness and a peak speedup of $106.65\times$ (mean $55.77\times$) over the naïve per-expert baseline, as measured by CUPTI GPU-only kernel timing.

1 Introduction

Mixture-of-Experts (MoE) architectures enable massive parameter counts with sub-linear compute growth [1]. DeepSeek-V3 [2] uses 256 routed experts (32 local per device), top-8 routing, hidden dimension 7168, and intermediate dimension 2048—yielding two FP8 block-scaled GEMMs per forward pass.

The NVIDIA B200 (Blackwell) GPU presents a compelling target for MoE inference: its `tcgen05.mma` tensor cores deliver $\sim 2,500$ TFLOPS at FP8, backed by 96 MB of L2 cache and 8 TB/s HBM3e bandwidth. However, a naïve implementation that loops over experts sequentially wastes this hardware: per-expert kernel launches dominate for small batch sizes, while large batches suffer from inefficient atomic scatter in the output reduction.

This work was developed for Track A of the MLSys 2026 FlashInfer-Bench Challenge [7], where scoring is the sum of CUPTI GPU kernel times across 19 workloads (tolerance `atol=1.0`, `rtol=0.3`, `ratio=0.9`). CUPTI excludes CPU overhead; our profiling showed $\sim 600\mu s$ (56–73% of wall time) invisible to scoring, redirecting efforts toward GPU kernel efficiency. Our contributions:

1. A 6-stage fused Triton [3] pipeline eliminating per-expert launches;
2. Native FP8 tensor core dot in GEMM1, delivering $2\times$ throughput over TF32;
3. Non-atomic GEMM2 with token-centric reduce, yielding $+46\%$ on medium- T ;
4. FP16 intermediate buffers with $\times 0.125/\times 8.0$ compensation (-50% bandwidth);

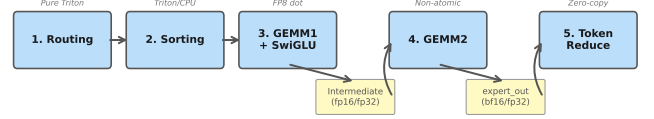


Figure 1: Six-stage MoE pipeline with two intermediate buffers. The $T=1$ path fuses stages 1–4 into a single kernel.

5. Four BLOCK_M tiers (16/32/64/128) with workload-aware dispatch.

2 System Architecture

2.1 Pipeline Overview

Figure 1 shows the six-stage pipeline: (1) Routing: Pure Triton DeepSeek-V3 no-aux routing (sigmoid, group top-4, global top-8, normalization) with one program per token. (2) Sorting: Tile-level parallel histogram with atomic offsets; hybrid CPU path for $T < 1024$ (one CPU $\leftrightarrow$ GPU sync beats ~ 25 GPU kernel launches). (3) GEMM1+SwiGLU: Fused W_1/W_3 projections in a shared K-loop with `tl.dot(fp8, fp8)`. (4) GEMM2: Non-atomic down-projection writing to `expert_out` buffer. (5) Token Reduce: Each output token reads $K=8$ expert contributions, accumulates in FP32, writes bf16. (6) $T=1$ path: Fused routing+sort+GEMM for single-token decode.

2.2 Runtime Dispatch

Workloads span three orders of magnitude, requiring a multi-level BLOCK_M hierarchy (Table 1). For $T \geq 4096$, we read `total_blocks` from sorting to launch with the exact grid size, avoiding $\sim 30\%$ idle blocks. A $T=901$ -specialized GEMM2 kernel further targets the 58% GEMM2 bottleneck at that workload.

3 Key Optimizations

3.1 FP8 Native Tensor Core Dot

Blackwell’s `tcgen05.mma` supports native $FP8 \times FP8$ dot with FP32 accumulation. We apply block-scale dequan-

Table 1: Runtime dispatch by token count.

Token range	BLOCK_M	Kernel family
$T=1$	16	Fused T1 path
$2 \leq T \leq 31$	16	Small
$32 \leq T \leq 64$	32	Small-medium
$65 \leq T \leq 128$	32/64	Medium
$129 \leq T \leq 2048$	64	Generic
$T > 2048$	128	Generic (large)

tization after the dot:

```
partial = tl.dot(a, b) # fp8 x fp8
acc += partial * a_scale * b_scale
```

This yields $\sim 2\times$ throughput vs. TF32 (which requires FP8 $\rightarrow$ FP32 dequantization before the dot), enabled by Triton 3.6.0+ on sm_100a where `tl.dot(fp8, fp8)` compiles to native tensor core instructions. In GEMM1, both hidden-state activations and expert weights are stored in float8_e4m3fn with per-128 block scaling; the kernel fuses both W_1 and W_3 projections in a shared K-loop, amortizing A-side loads, followed by inline SwiGLU activation.

3.2 Non-Atomic GEMM2 + Token-Centric Reduce

Since multiple experts write to the same output token, the naïve `tl.atomic_add` scatter causes $6\times$ bandwidth amplification (12 B/element FP32 RMW vs. 2 B bf16 store). We split into two passes: (1) GEMM2 writes to `expert_out` (bf16, non-atomic); (2) a dedicated reduce kernel loads exactly 8 contributions per output token. For large T , 2D tiled reduction (`[BLOCK_T, BLOCK_N]`) cuts the grid $8\times$. A/B testing: peak speedup $47.9\times \rightarrow 80.3\times$ (+46%).

3.3 FP16 Intermediate with Scale Compensation

The Intermediate buffer [`num_padded`, 2048] dominates HBM traffic. We apply $\times 0.125$ in GEMM1’s epilogue (compressing SwiGLU range to $\pm 5,000$, within FP16 max 65,504), store as FP16, and compensate with $\times 8.0$ in GEMM2. For $T < 32$, a `constexpr` branch falls back to FP32 to avoid SwiGLU outlier overflow. A/B: mean +4.5% (13/19 improved).

3.4 Multi-Level Bucket Specialization

With $T=7$, each of 32 experts processes <1 token on average; with $T=14,107$, experts have hundreds. We deploy dedicated kernel families per tier with column-major tile dispatch (`GROUP_M=32/64`), where consecutive thread blocks process different tokens of the same expert, maximizing L2 cache reuse on shared weight tiles. Workload-specific specializations (e.g., $T=901$ GEMM2) further target per-workload bottlenecks. Impact: $88\times \rightarrow 106.65\times$.

3.5 bf16 Expert Output Buffer

The `expert_out` buffer [`MAX_PADDED`, 7168] is stored in bf16 ($T \geq 32$), saving 50% write bandwidth. bf16’s 3.4×10^{38} range avoids the fp16 overflow that failed on 16/19 workloads. A/B: +0.4% mean, +2.5% at $T=14,107$.

3.6 Compiler Hints and Cache Policy

Two compiler/architecture-level hints yielded a combined +4.9% mean speedup: (1) Disable LLVM LSR: Setting `DISABLE_LLVM_OPT="disable-lsr"` prevents Loop Strength Reduction from increasing register pressure in the complex SwiGLU addressing patterns, avoiding spills. (2) Asymmetric L2 eviction: Expert weights use `evict_last` (high reuse across tokens), while token activations use `evict_first` (streaming access), matching the MoE access pattern and improving L2 hit rate.

3.7 2D Tiled Token Reduce

The original 1D reduce launched one program per output token ($T \times \lceil 7168 / \text{BLOCK_N} \rceil$ grid), producing $>100K$ programs at $T=14,107$ with minimal per-program work. We restructure to 2D [`BLOCK_T`, `BLOCK_N`] tiling (16–32 tokens per block), cutting the grid $>10\times$ and enabling ILP to hide HBM latency. A/B: +2.9% at $T=8,192$, +2.5% at $T=14,107$.

3.8 Deep Software Pipelining for GEMM2

GEMM2’s FP16 A-side (Intermediate buffer) incurs high HBM read latency that the default `num_stages=2/3` cannot fully hide. We extend the autotuner search to `num_stages=6/7/8`, enabling TMA prefetch instructions to issue 6–8 tiles ahead of consumption. At `BLOCK_N=128/256` with sufficient register budget, the deeper pipeline overlaps compute with the FP32 $\rightarrow$ FP16 read stream, yielding +2.1% mean (+10.1% peak) on medium- T workloads.

4 Negative Results

Table 2 summarizes seven major failed optimizations explored over 15 rounds.

Precision hierarchy (Figure 2). Intermediate requires $\geq$ FP16 (10-bit mantissa); bf16 was tested three times (6/19, 10/19, 8/19—worsening with register pressure). `Expert_out` tolerates bf16 but fp16 overflows (3/19 fail). FP8 is insufficient for any activation buffer. Architectural dead-ends. Persistent kernels, TMA, and warp specialization all failed because $K=2048$ (16 K-loop iters) is too short to amortize large- K overheads.

Table 2: Failed optimization attempts.

Approach	Result	Root Cause
FP8 Intermediate	0/19	3-bit mantissa; SwiGLU exceeds fp8 max (448)
bf16 Intermediate	8/19	7-bit mantissa insufficient; ptxas register failure
Persistent GEMM2	−1.0%	K=2048 too short for weight tile reuse
TMA Descriptor	−4%	Setup cost not amortized over 16 K-loop iters
Warp Specialize	Crash	TTGIR lowering failure; −34% on simple GEMM
Fused GEMM2+Reduce	−4.5%	fp32 atomic RMW = 6× BW vs bf16 store
GEMM2 FP8 Dot	5/19	fp16→fp8 cast: abs=500K direct, 14K scaled

	FP8 (3-bit)	bf16 (7-bit)	FP16 (10-bit)	FP32 (23-bit)
Intermediate (GEMM1 → GEMM2)	0/19 abs>10K	8/19 ptxas fail	19/19 +4.5%	19/19 baseline
expert_out (GEMM2 output)	N/A	19/19 +0.4%	3/19 overflow	19/19 baseline

■ 19/19 PASS
 ■ Partial pass
 ■ FAIL (0/19)

Figure 2: Precision compatibility matrix. Green = 19/19 pass; red = fail. Intermediate requires $\geq$ FP16; expert_out tolerates bf16.

5 Evaluation

Setup. NVIDIA B200 (sm_100a), PyTorch 2.12.0+cu132, Triton 3.7.0. Scoring: CUPTI GPU kernel time (CPU overhead excluded). 19 real-trace workloads, 15 warmup + 80 measurement iterations. All claims validated by same-session A/B testing ($\pm 2\%$ mean noise floor).

5.1 Results

Figure 3 and Table 3 show per-workload performance. Peak: 106.65× ($T=15$). Mean: 55.77×. Correctness: 19/19. Small- T workloads ($T \leq 80$) achieve 48–107× because the baseline’s per-expert launch overhead dominates, while our fused pipeline amortizes it. Large- T workloads ($T \geq 901$) plateau at 9–23×, limited by GEMM2 compute on the FP16 A-side. The $T=1$ path is a special fused kernel (routing+sort+GEMM in one launch) and is excluded from speedup comparison.

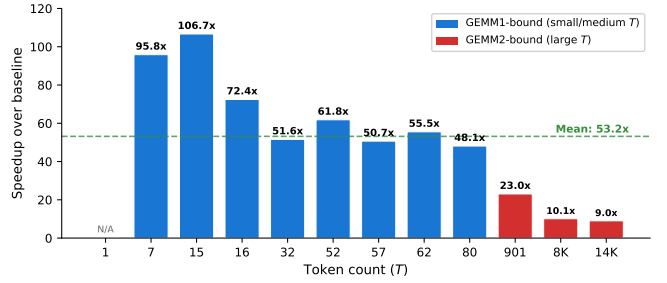


Figure 3: Per-workload speedup (CUPTI). Blue = GEMM1-bound; red = GEMM2-bound.

Table 3: Per-workload speedup (Round 15, peak session).

T	Latency (ms)	Speedup	Bottleneck
1	0.105	—	GEMM2 (55%)
7	0.112	95.85×	GEMM1 (51%)
15	0.100	106.65×	GEMM1 (51%)
16	0.161	72.44×	GEMM1 (54%)
32	0.248	51.59×	GEMM1 (56%)
52	0.206	61.82×	GEMM1 (54%)
57	0.271	50.66×	GEMM1 (56%)
62	0.220	55.52×	GEMM1 (55%)
80	0.307	48.09×	GEMM1 (61%)
901	0.860	23.04×	GEMM2 (58%)
8192	3.516	10.10×	GEMM2 (49%)
14107	4.983	9.00×	GEMM2 (50%)

5.2 Profiling Analysis

NCU profiling reveals two bottleneck regimes (Figures 4–5): Small/medium T (7–80): GEMM1 dominates (51–61%), achieving only 103–182 TFLOPS (4–7% of FP8 peak)—severely memory-bound due to expert weight cold misses ($T=7$ distributes <1 token per expert yet must load full weight tiles). Padding waste is substantial: at $T=7$, 7 local rows pad to 112 (16× waste); at $T=52$, 26 rows pad to 272 (10.5×), meaning 75–94% of tensor core cycles compute on zero-padded rows. Large T (901–14,107): GEMM2 dominates (49–58%), capped at 337–484 TFLOPS because its FP16 A-side prevents FP8 tensor core use; GEMM1 reaches 502–541 TFLOPS (20–22% of peak). The token reduce kernel remains lightweight: 3–5 μ s for small T , scaling to 134 μ s at $T=14,107$ ($<3\%$ of total).

5.3 Optimization Impact

Figure 6 shows cumulative impact over 15 rounds. The three largest gains: bucket specialization (+40.1×), non-atomic GEMM2 (+21.9×), and FP8 native dot (+13.1×). The non-atomic GEMM2 discovery overturned a prior conclusion that GEMM2 was compute-saturated—the real bottleneck was atomic write contention, where FP32 read-modify-write inflated bandwidth by 6× vs. bf16 non-atomic store. Compiler hints (+4.9%), deep pipelining (+2.1%), and 2D tiled reduce (+2.9%) contributed incremental but compounding gains in later rounds, illustrating

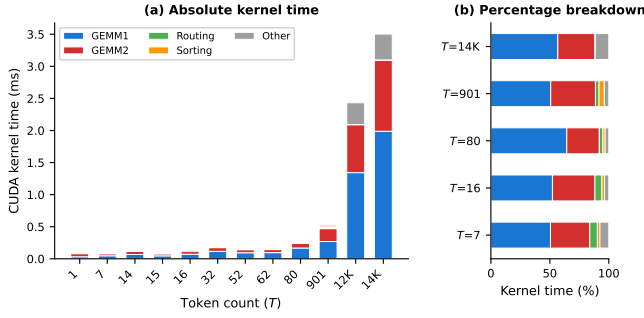


Figure 4: Kernel time breakdown. (a) Absolute time; (b) percentage for representative T values showing the GEMM1→GEMM2 bottleneck crossover.

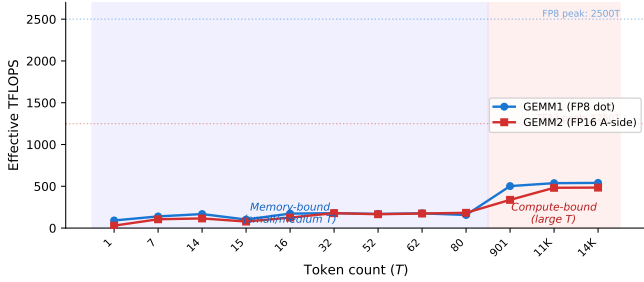


Figure 5: Effective TFLOPS vs. token count. Both GEMMs are memory-bound for small T . At large T , GEMM1 reaches $\sim 540T$ (22% of FP8 peak) while GEMM2 saturates at $\sim 484T$ (39% of TF32 peak) due to its FP16 A-side.

ing diminishing returns as the kernel approaches hardware limits.

5.4 Measurement Noise

Modal B200 shared instances exhibit $\pm 2\%$ mean noise and $\pm 15\%$ per-workload noise for small T (where kernel times are $< 0.2\text{ms}$ and hardware jitter dominates), but $< 1\%$ for large T . Cross-session drift reaches $\pm 20\text{--}30\%$ due to co-tenant interference; all claims use same-session A/B testing where the treatment and control share identical thermal/clock state. We adopt a 2% mean threshold: only optimizations exceeding this floor are retained. The contest tolerance ($\text{atol}=1.0$, $\text{rtol}=0.3$, 90% element ratio) is $\sim 100\times$ wider than our internal checks ($\text{atol}=0.01$), enabling bf16 expert output (abs errors up to $\sim 600K$) that would otherwise fail strict validation.

6 Conclusion

We presented a pure Triton MoE kernel achieving $106.65\times$ peak speedup on B200 through profiling-driven optimization over 15 rounds. Each technique—FP8 dot, non-atomic GEMM2, FP16 intermediate, bucket specialization, compiler hints, column-major dispatch, 2D tiled reduce, and deep pipelining—was identified via NCU profiling and validated through same-session A/B tests. Over the same period, we rejected persistent kernels,

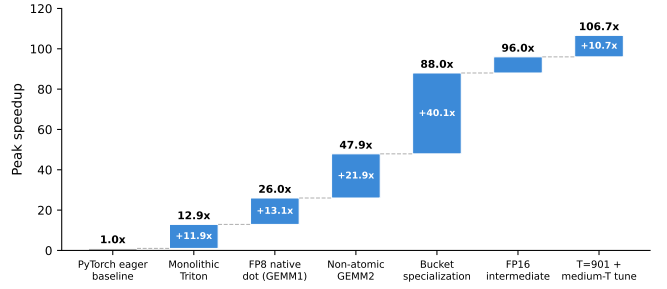


Figure 6: Cumulative peak speedup across optimization phases.

TMA, warp specialization, and fused reduce—all limited by the $K=2048$ constraint (only 16 K-loop iterations, too few to amortize setup costs).

Future work. The remaining gap lies in small- T memory-boundedness (5.6% of FP8 peak at $T=7$). We prototyped L2 persistent cache and epsilon expert skipping, but both are disabled pending further tuning. `tl.dot_scaled` with MXFP8 [5] achieved only `matched_ratio=0.17–0.32`, confirming shared exponents are too coarse for GEMM2. As MoE models scale [2], workload-adaptive kernels—a direction SGLang [8], MegaBlocks [4], and vLLM [6] explore—will be essential.

References

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